

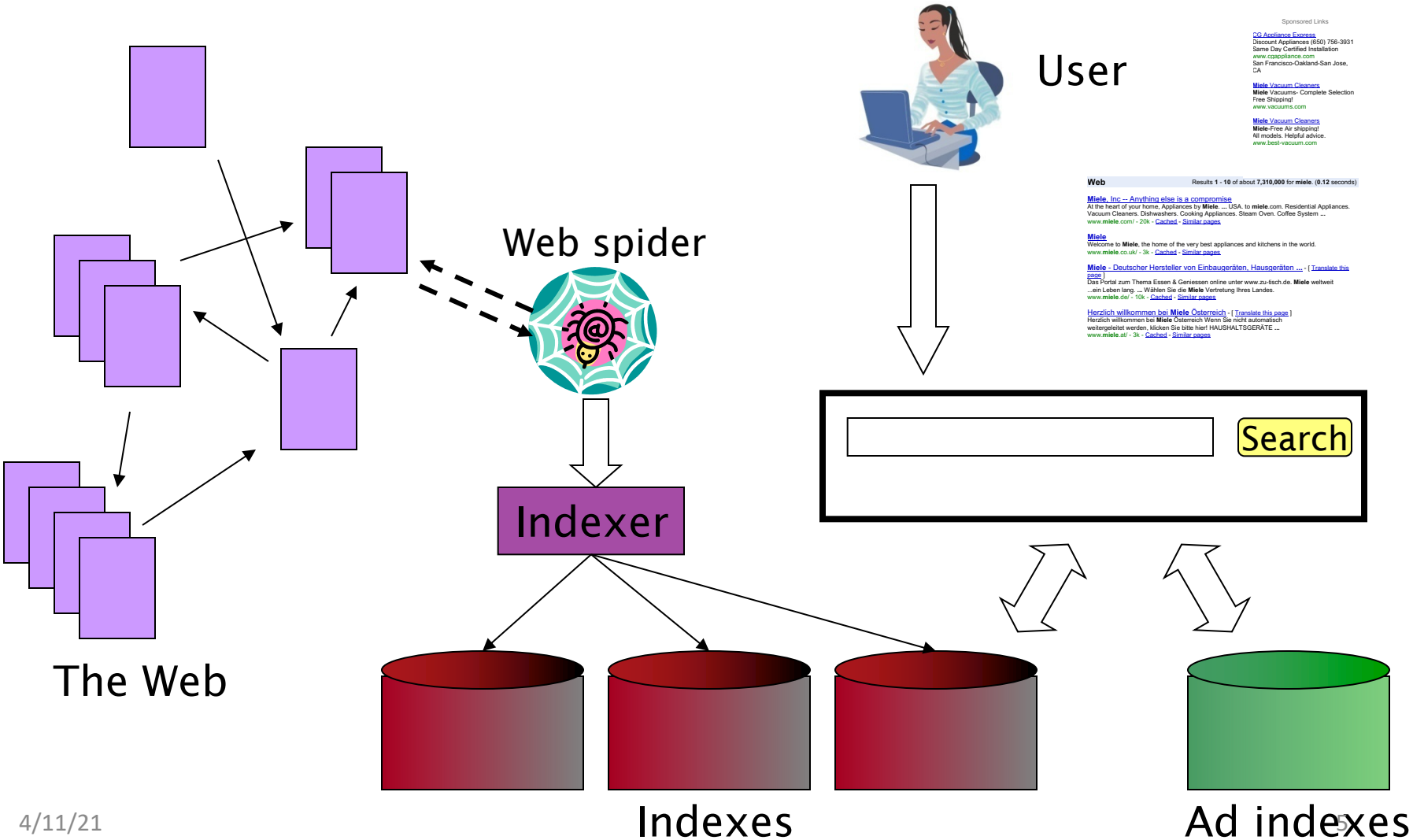
Introduction to **Information Retrieval**

Review

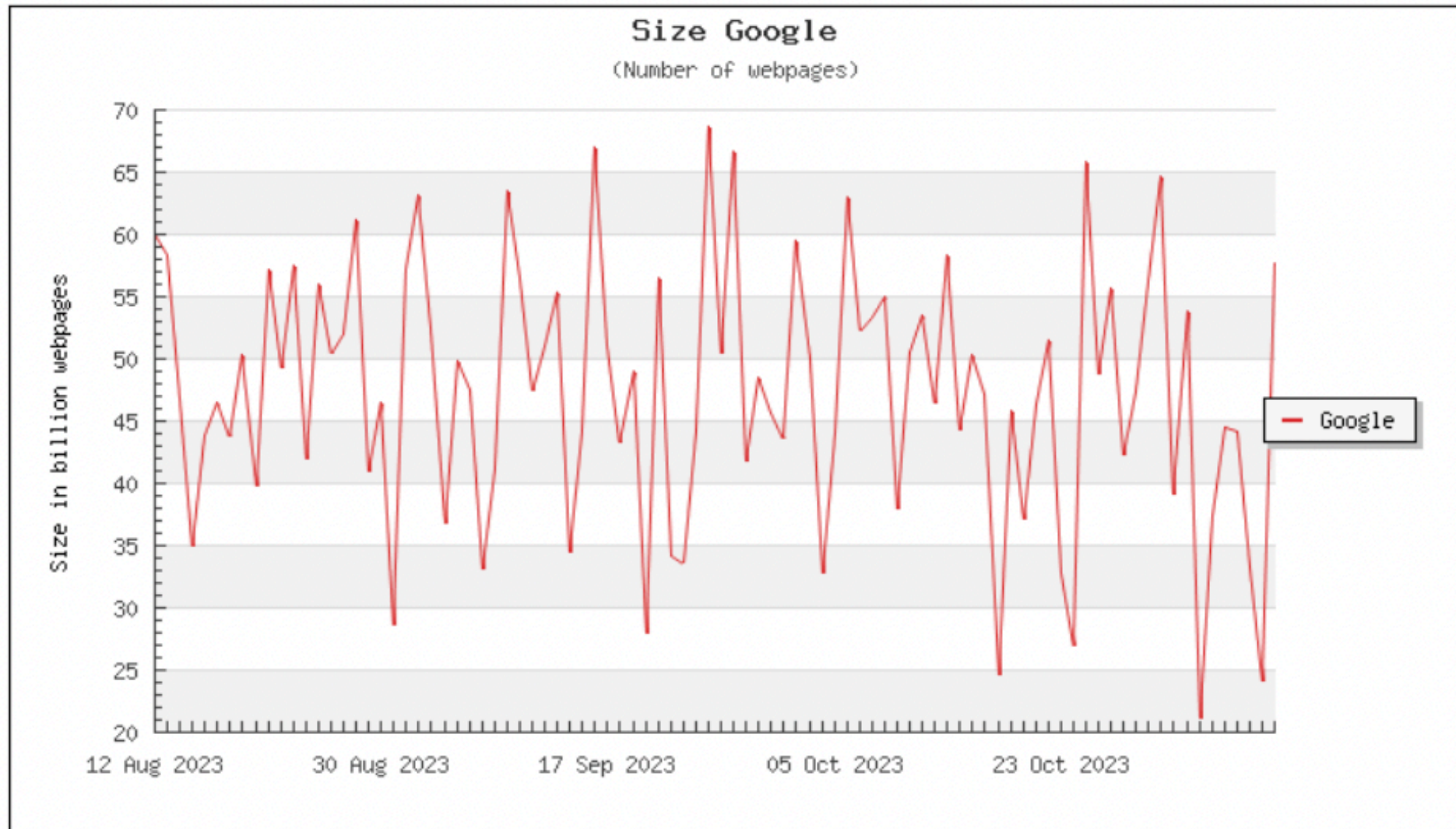
Agenda

- Project
- Lecture revisions
- Final exam
- Announcements

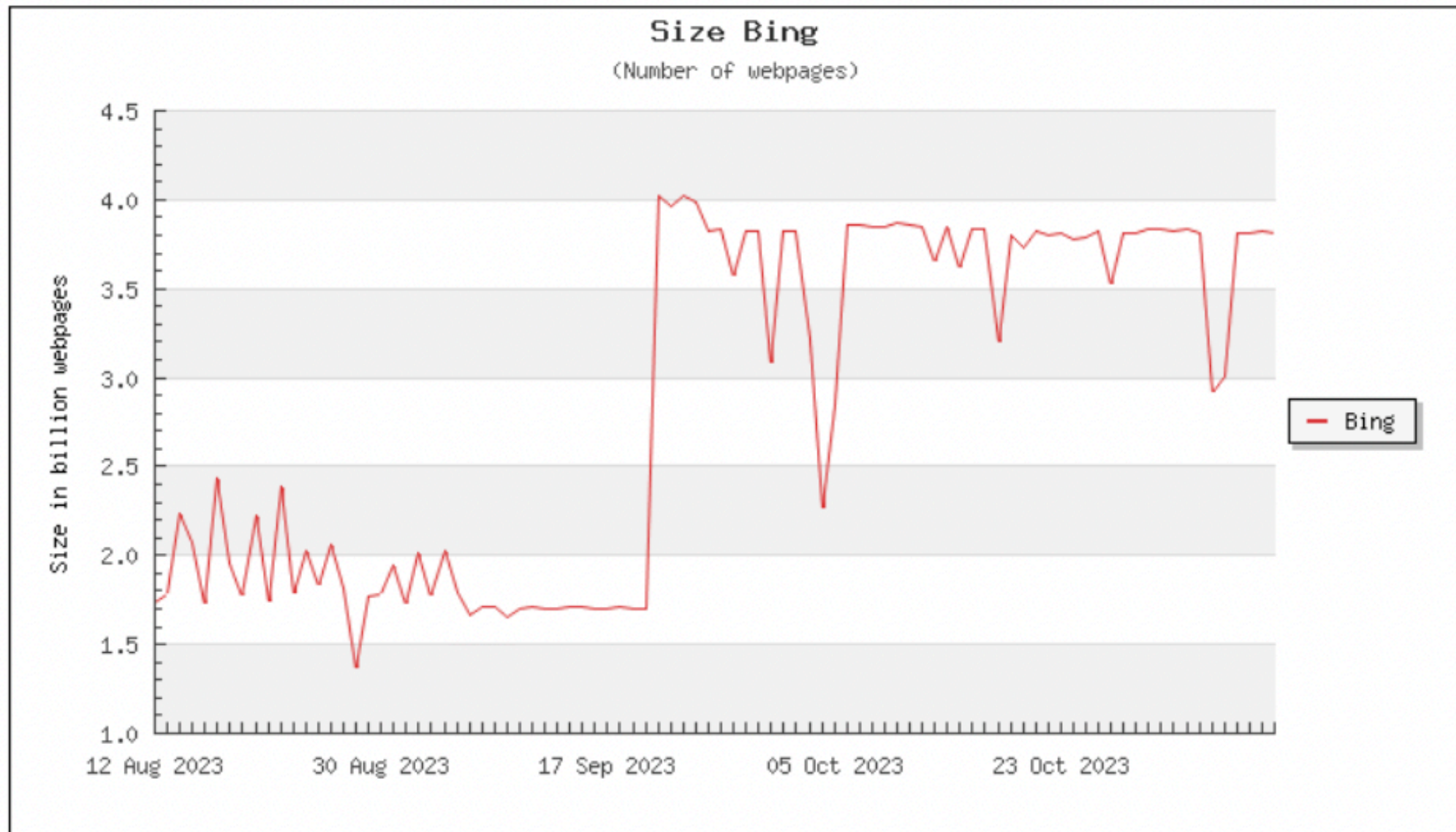
Web search basics



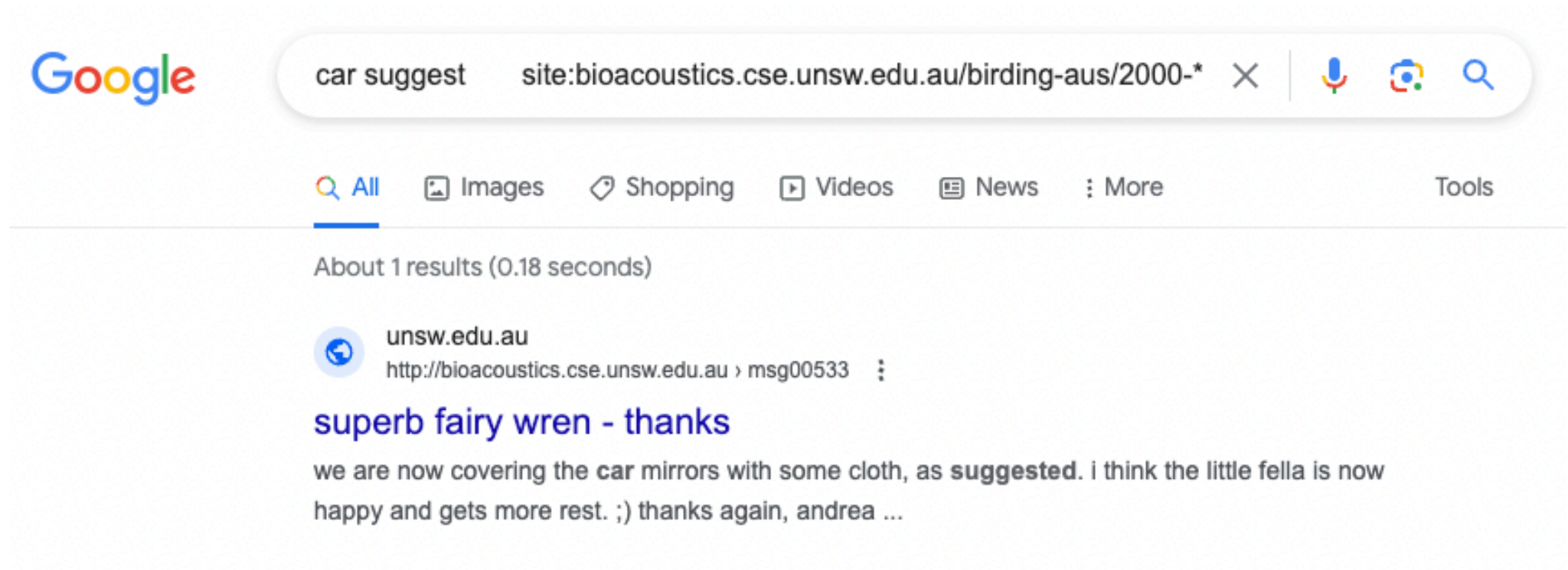
Project related



Project related



Project related



The screenshot shows a Google search interface. The search bar contains the text "car suggest" followed by a filter "site:bioacoustics.cse.unsw.edu.au/birding-aus/2000-*". Below the search bar, there are tabs for "All", "Images", "Shopping", "Videos", "News", and "More". The "All" tab is selected. Below the tabs, it says "About 1 results (0.18 seconds)". The search result is from "unsw.edu.au" and the URL is "http://bioacoustics.cse.unsw.edu.au › msg00533". The title of the result is "superb fairy wren - thanks" in blue. The snippet of the result reads: "we are now covering the car mirrors with some cloth, as suggested. i think the little fella is now happy and gets more rest. ;) thanks again, andrea ...".

Google

car suggest site:bioacoustics.cse.unsw.edu.au/birding-aus/2000-*

All Images Shopping Videos News More Tools

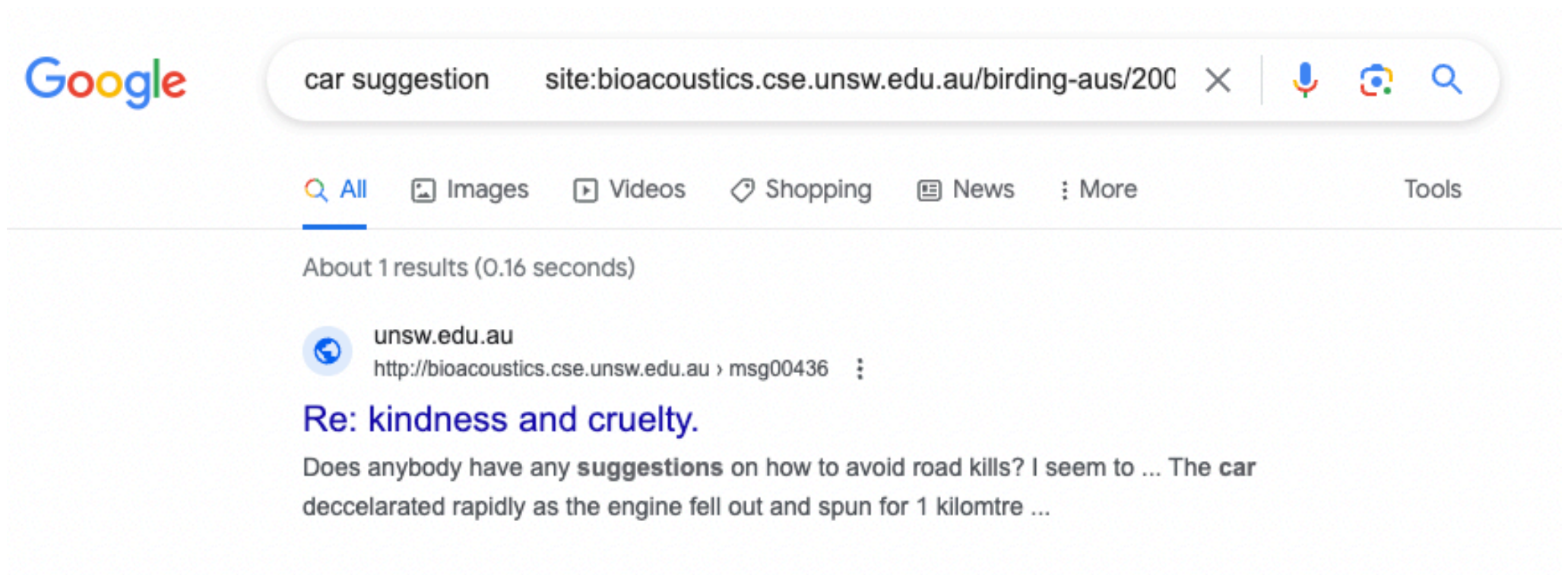
About 1 results (0.18 seconds)

unsw.edu.au
http://bioacoustics.cse.unsw.edu.au › msg00533

superb fairy wren - thanks

we are now covering the car mirrors with some cloth, as **suggested**. i think the little fella is now happy and gets more rest. ;) thanks again, andrea ...

Project related



The screenshot shows a Google search interface. The search bar contains the text "car suggestion" followed by "site:bioacoustics.cse.unsw.edu.au/birding-aus/200". Below the search bar, the "All" tab is selected. The search results show "About 1 results (0.16 seconds)". The single result is from "unsw.edu.au" with the URL "http://bioacoustics.cse.unsw.edu.au › msg00436". The result title is "Re: kindness and cruelty." and the snippet reads: "Does anybody have any **suggestions** on how to avoid road kills? I seem to ... The car decclarated rapidly as the engine fell out and spun for 1 kilomtre ...".

Google

car suggestion site:bioacoustics.cse.unsw.edu.au/birding-aus/200

All Images Videos Shopping News More Tools

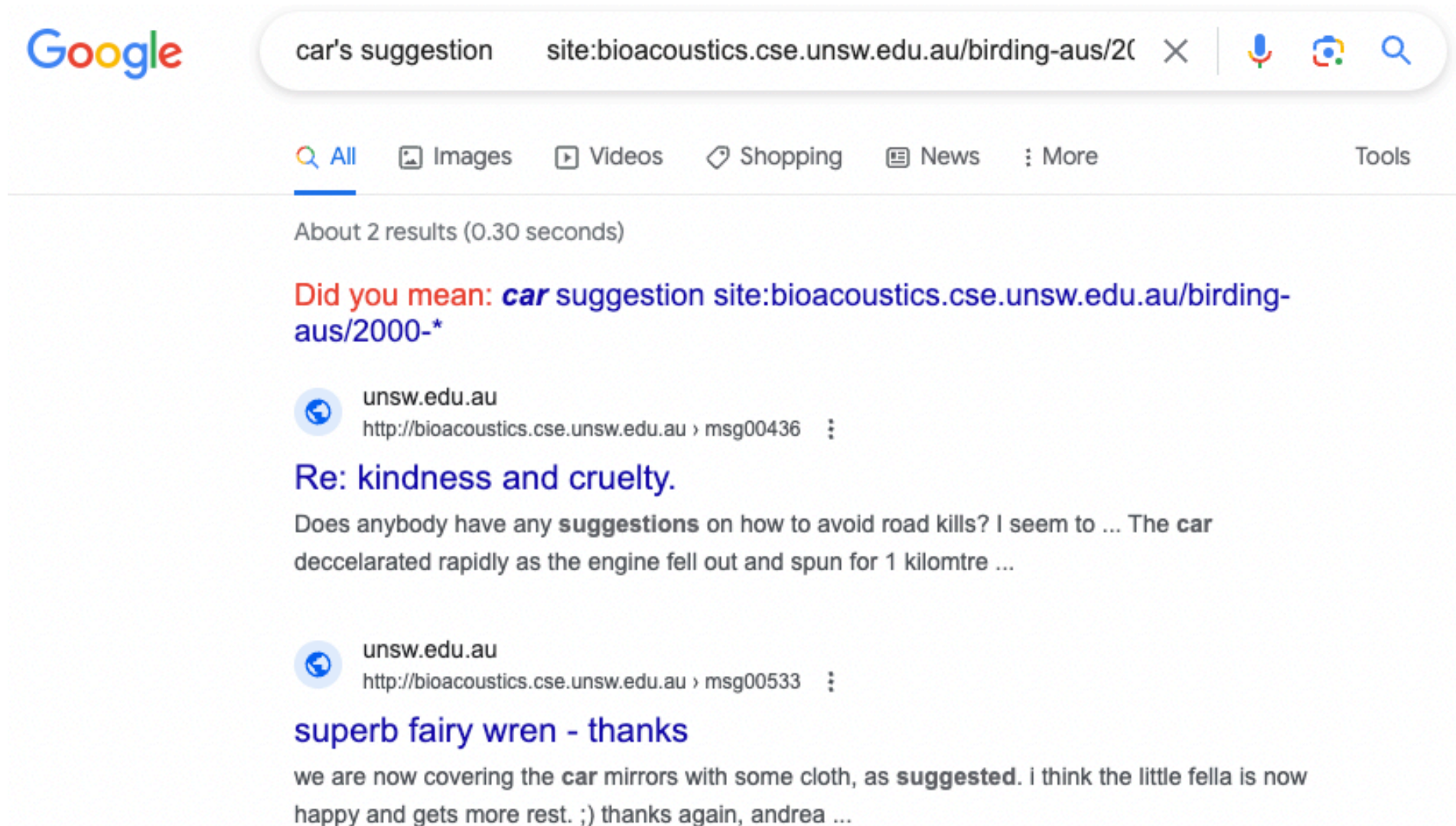
About 1 results (0.16 seconds)

unsw.edu.au
http://bioacoustics.cse.unsw.edu.au › msg00436

Re: kindness and cruelty.

Does anybody have any **suggestions** on how to avoid road kills? I seem to ... The car decclarated rapidly as the engine fell out and spun for 1 kilomtre ...

Project related (issue)



The screenshot shows a Google search interface. The search bar contains the text "car's suggestion site:bioacoustics.cse.unsw.edu.au/birding-aus/2000-". Below the search bar, the Google logo is on the left, and navigation links for "All", "Images", "Videos", "Shopping", "News", and "More" are in the center. The "All" link is selected. On the right, there are icons for voice search, image search, and a magnifying glass. Below the navigation bar, it says "About 2 results (0.30 seconds)". The first result is a snippet from "unsw.edu.au" with the URL "http://bioacoustics.cse.unsw.edu.au › msg00436". The title is "Re: kindness and cruelty." and the text says "Does anybody have any **suggestions** on how to avoid road kills? I seem to ... The car decclarated rapidly as the engine fell out and spun for 1 kilomtre ...". The second result is also from "unsw.edu.au" with the URL "http://bioacoustics.cse.unsw.edu.au › msg00533". The title is "superb fairy wren - thanks" and the text says "we are now covering the **car** mirrors with some cloth, as **suggested**. I think the little fella is now happy and gets more rest. ;) thanks again, andrea ...".

Google

car's suggestion site:bioacoustics.cse.unsw.edu.au/birding-aus/2000-

Q All Images Videos Shopping News More Tools

About 2 results (0.30 seconds)

Did you mean: **car** suggestion site:bioacoustics.cse.unsw.edu.au/birding-
aus/2000-*

unsw.edu.au
http://bioacoustics.cse.unsw.edu.au › msg00436

Re: kindness and cruelty.

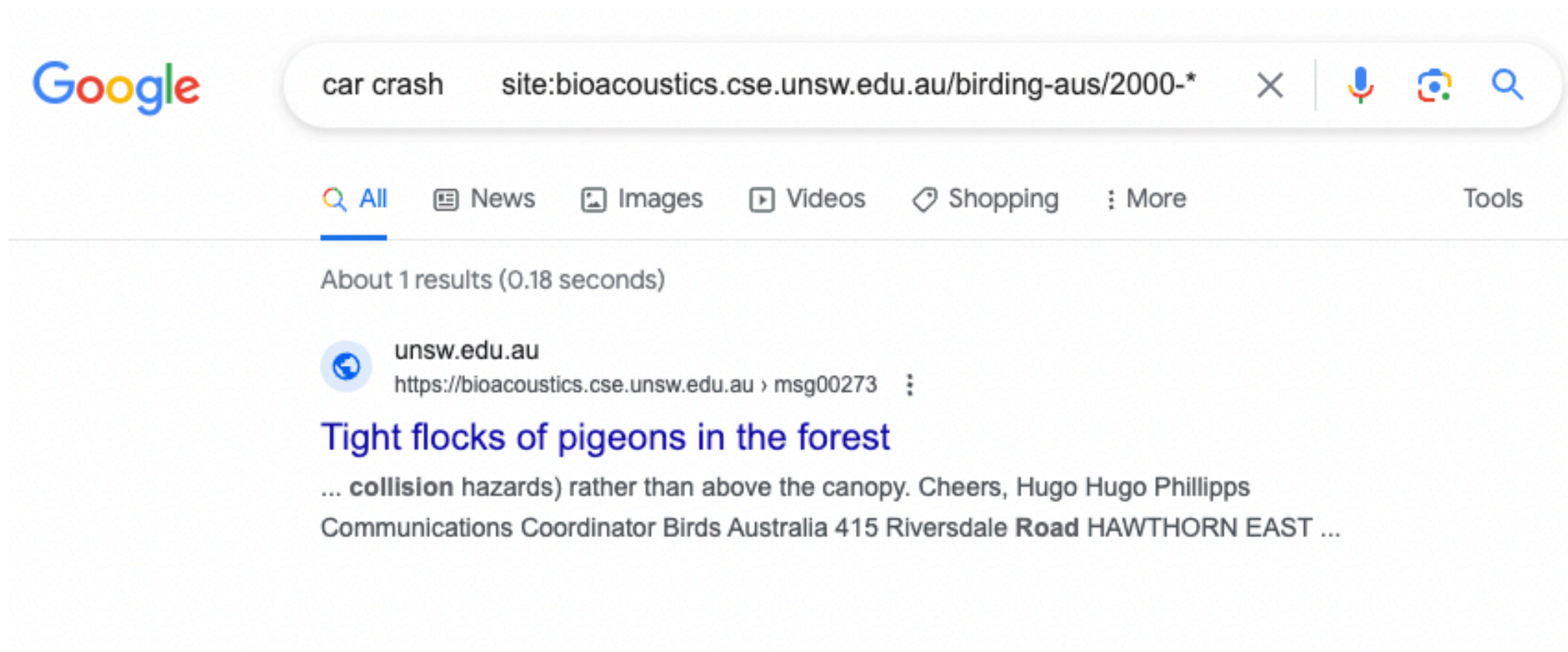
Does anybody have any **suggestions** on how to avoid road kills? I seem to ... The car
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happy and gets more rest. ;) thanks again, andrea ...

Project related (issue)



The screenshot shows a Google search interface. The search bar contains the text "car crash site:bioacoustics.cse.unsw.edu.au/birding-aus/2000-*". Below the search bar, the "All" tab is selected. The search results show "About 1 results (0.18 seconds)". The first result is from "unsw.edu.au" with the URL "https://bioacoustics.cse.unsw.edu.au › msg00273". The title of the result is "Tight flocks of pigeons in the forest". The snippet of the result is "... collision hazards) rather than above the canopy. Cheers, Hugo Hugo Phillipps Communications Coordinator Birds Australia 415 Riversdale Road HAWTHORN EAST ...".

Google

car crash site:bioacoustics.cse.unsw.edu.au/birding-aus/2000-*

Q All News Images Videos Shopping More Tools

About 1 results (0.18 seconds)

unsw.edu.au
https://bioacoustics.cse.unsw.edu.au › msg00273

Tight flocks of pigeons in the forest

... collision hazards) rather than above the canopy. Cheers, Hugo Hugo Phillipps
Communications Coordinator Birds Australia 415 Riversdale Road HAWTHORN EAST ...

Lecture revisions

Boolean Model

- incidence vector/matrix
- semantics of the query model (AND/OR/NOT, and other operators, e.g., /k, /S)
- inverted index, positional inverted index
- query processing methods for basic and advanced boolean queries (including phrase query, queries with /S operator, etc.)
- query optimization methods (list merge order, skip pointers)

Preprocessing

- typical preprocessing steps: tokenization, stopword removal, stemming/lemmatization,

Tolerant Retrieval

- Wildcard queries
- Permuterm index
- Bigram index
- Spelling correction
- Noise channel model

Other Topics

- Soundex
- N-grams
- Maximum Likelihood Estimate

Index Construction

- Why we need dedicated algorithms to build the index?
- BSBI: Blocked sort-based indexing
- SPIMI: Single-pass in-memory indexing
- Dynamic indexing: Immediate merge, no merge, logarithmic merge

Index Compression

- Heap's law, Zipf's law
- Dictionary compression
 - Dictionary as a string
 - Front encoding
- posting lists compression
 - Elias encoding
 - Variable length encoding
- Shannon limit

Vector Space Model

- What is/why ranked retrieval?
- raw and normalized tf, idf
- cosine similarity
- tf-idf variants (using SMART notation): e.g., Inc.ltc
- basic query processing method: document-at-a-time vs term-at-a-time
- exact & approximate query optimization methods (heap-based top-k algorithm, MaxScore algorithms, etc.)
- Query processing methods based on advanced or tiered inverted indexes (e.g., high/low lists)

tf-idf example: Inc.Itc

Document: *car insurance auto insurance*

Query: *best car insurance*

Term	Query						Document				Prod
	tf-raw	tf-wt	df	idf	wt	n'lize	tf-raw	tf-wt	wt	n'lize	
auto	0	0	5000	2.3	0	0	1	1	1	0.52	0
best	1	1	50000	1.3	1.3	0.34	0	0	0	0	0
car	1	1	10000	2.0	2.0	0.52	1	1	1	0.52	0.27
insurance	1	1	1000	3.0	3.0	0.78	2	1.3	1.3	0.68	0.53

Exercise: what is N , the number of docs?

$$\text{Doc length} = \sqrt{1^2 + 0^2 + 1^2 + 1.3^2} \approx 1.92$$

$$\text{Score} = 0 + 0 + 0.27 + 0.53 = 0.8$$

Cosine for length-normalized vectors

- For length-normalized vectors, cosine similarity is simply the dot product (or scalar product):

$$\cos(\vec{q}, \vec{d}) = \vec{q} \bullet \vec{d} = \sum_{i=1}^{|V|} q_i d_i$$

for q, d length-normalized.

Evaluation

- Existing method to prepare for the benchmark dataset, queries, and ground truth
- For unranked results: Precision, recall, F-measure
- For ranked results: precision-recall graph, 11-point interpolated
- precision, MAP, etc.

Web Search Basics

- Estimation of relative sizes of two search engines.
- Near duplicate detection: the shingling method

Crawling

- Understand the requirements of crawlers
- Mercator scheme
- **Not required:** optimization for age

Link Analysis

- The pagerank algorithm: theory and practice

COMP6714 vs COMP9319

- COMP6714 – Information Retrieval
 - Search basics, search quality
 - Python
- COMP9319 – Web Data Compression and Search
 - Size, performance & scalability
 - C/C++

Week	Topic
1	Course Introduction + Boolean Retrieval
2	Preprocessing
3	Index construction
4	Compression
5	Vector Space Model
6	Flexibility Week (no lecture)
7	Evaluation
8	Crawling
9	Link Analysis
10	Revision and Exam Preparation

Assessment

```
asgt      = mark for the assignment      (out of 10)
proj      = mark for the project          (out of 40)
exam      = mark for final exam          (out of 50)
okEach    = exam > 20                    (after scaling)
mark      = asgt + proj + exam
grade     = HD|DN|CR|PS  if mark >= 50 && okEach
           = FL          if mark < 50 && okEach
           = UF          if !okEach
```

Final exam (2hrs, worth 50%)

- Exam date: In-person, 30 Nov (Thu) afternoon, 1:45pm-4:00pm
- Exam instructions will be available 15mins before 2:00pm
- If you are ill on the day of the exam, do not attend the exam – c.f. UNSW fit-to-sit policy. Apply for special consideration asap.
- If granted by UNSW SC, supp exam is likely to be of different format, e.g., long questions / programming / oral exam / mixed

Final exam format

- 2 hours exam, closed book exam
- You may refer to the lecture slides
- Bring a **UNSW-approved calculator & pens**
- A programming question + M.C. / written questions
- Similar “style” as your assignment but of a larger scope

Access to Python documentation

Python 3.9.17 documentation

Welcome! This is the official documentation for Python 3.9.17.

Parts of the documentation:

[What's new in Python 3.9?](#)

or all "What's new" documents since 2.0

[Tutorial](#)

start here

[Library Reference](#)

keep this under your pillow

[Language Reference](#)

describes syntax and language elements

[Python Setup and Usage](#)

how to use Python on different platforms

[Python HOWTOs](#)

in-depth documents on specific topics

[Installing Python Modules](#)

installing from the Python Package Index & other sources

[Distributing Python Modules](#)

publishing modules for installation by others

[Extending and Embedding](#)

tutorial for C/C++ programmers

[Python/C API](#)

reference for C/C++ programmers

[FAQs](#)

frequently asked questions (with answers!)

Indices and tables:

[Global Module Index](#)

quick access to all modules

[General Index](#)

all functions, classes, terms

[Glossary](#)

[Search page](#)

search this documentation

[Complete Table of Contents](#)

lists all sections and subsections

How to prepare for the exam...

COMP6714 Final Exam

Exam Paper

Analogue Clock

Digital Clock

Calculator (GNOME)

-- Terminals -----

XTerm (short)

XTerm (tall)

XTerm (inverse)

XTerm (big font)

-- Editors -----

gedit

nedit

nedit (big font)

vscode

-- Keyboard Layouts -----

us keyboard

us.dvorak keyboard

us.colemak keyboard

-- Screen Layouts -----

Screen Resolutions

Magnifier

Logout

COMP6714 23T3 - Final Exam — Mozilla Firefox

COMP6714 23T3 - Final Exam x COMP6714 23T3 - Lecture Six +

file:///home/class/public_html/index.html

COMP6714 23T3 Final Exam

Thursday 30th November 2023

Marks: 100

Time: 10 minutes reading + 2 hours working

Please read all of the instructions below while you are waiting.

Instructions

Start-of-exam Instructions

- Place your student card/other photo ID on the desk.
- Wait for the supervisor to announce the start of reading time.
- Once reading time has started, click the button below to reveal the questions.

End-of-exam Instructions

- Stop working when your supervisor tells you to do so.

The screenshot shows a web browser window with the title "COMP6714 23T3 Final Exam". The address bar shows the URL "http://home/class/public_html/index.html". The page content includes the following text:

COMP6714 23T3 Final Exam

Thursday 30th November 2023

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A terminal window is open in the foreground, showing the prompt "[vong]@vx05:~\$".

A sample question

Question 7 (3 marks)

Write your answers for this question in **q7.txt**.

Which of the following are queries that skip pointers are useful for?

A	x OR y
B	x AND y
C	x OR (NOT y)
D	x AND (NOT y)
E	NOT x

Submission

Submit using the following command:

```
$ submit q7
```

q7.txt

q7.txt

x

Question 7

Please replace "X, Y, Z" inside the square brackets with your answer, e.g., "A, B". Do not remove the square brackets.

ANSWER: [X, Y, Z]

q7.txt

q7.txt

Question 7

Please replace "X, Y, Z" inside the square brackets with your answer, e.g., "A, B". Do not remove the square brackets.

ANSWER: [B, D]

Another sample question

Question 8 (6 marks)

Write your answers for this question in **q8.txt**.

Suppose a search engine has indexed totally 10 documents, with 3 of them containing a keyword Apple.

A search on Apple returns 5 documents. Of the 5 documents retrieved, 3 contains Apple. Its results are as follows (the leftmost item is the top ranked search result):

A A N A N

where A and N represent documents containing Apple and those without Apple respectively.

Based on the above query, what is the MAP (mean average precision) of the search engine?

Round your answer to two decimal places.

Submission

Submit using the following command:

```
$ submit q8
```

q8.txt

q8.txt

×

Question 8

Please replace "X" inside the square brackets with your answer. Do not remove the square brackets.

ANSWER: [X]

q8.txt

q8.txt

×

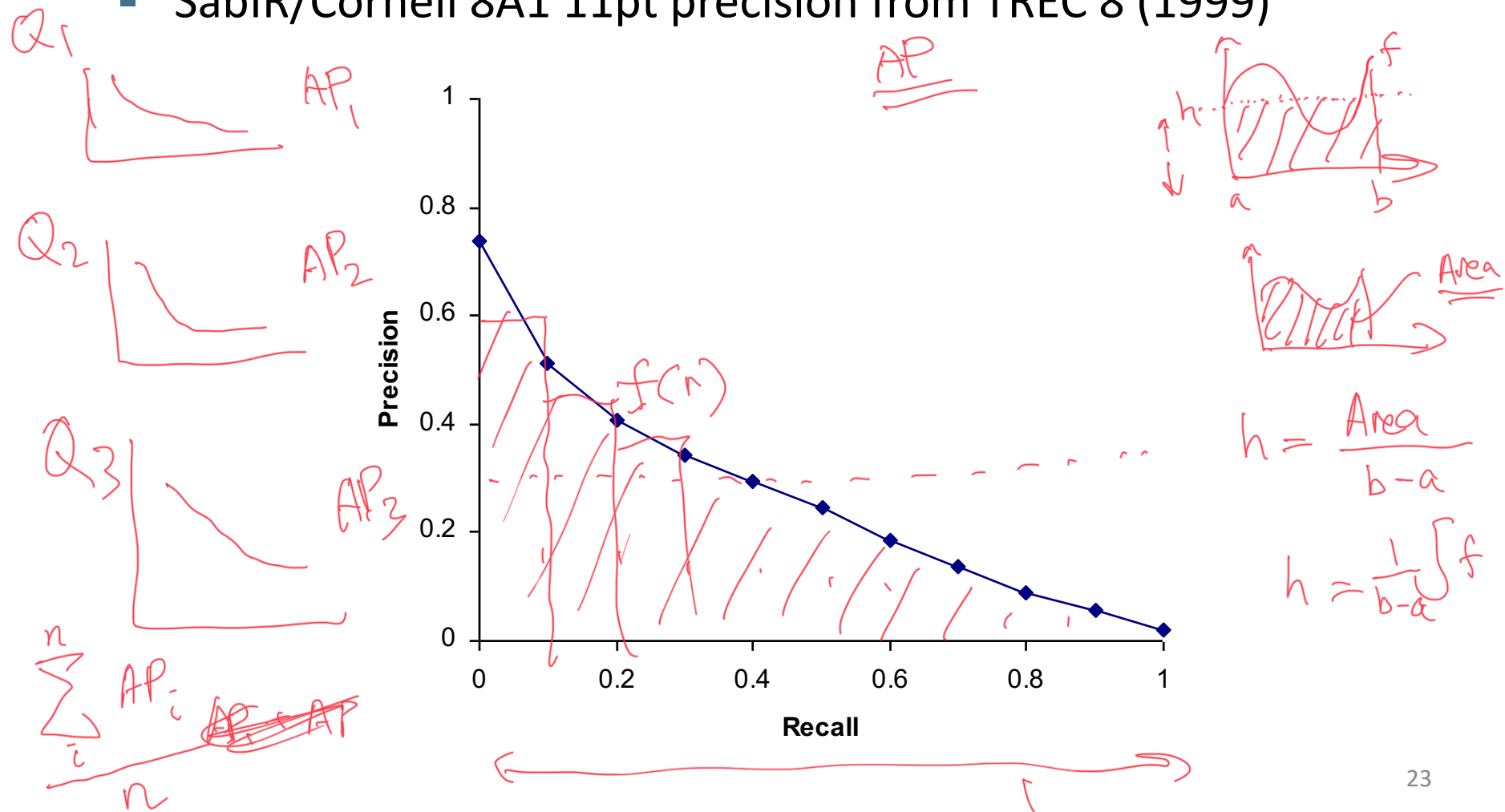
Question 8

Please replace "X" inside the square brackets with your answer. Do not remove the square brackets.

ANSWER: [0.92]

Typical (good) 11 point precisions

- SabIR/Cornell 8A1 11pt precision from TREC 8 (1999)



Q8

	A	B	C	D	E	F
	Table 1					
1						
2		prec@1	prec@2	prec@3	prec@4	prec@5
3		1	1	0.67	0.75	0.6
4	Recall	0.33	0.67	0.67	1	1
5						
6						
7		0.916666666666667				
8						
9						
Formula		(B3 + C3 + E3) ÷ 3				

Yet more evaluation measures...

- Mean average precision (MAP)
 - Average of the precision value obtained for the top k documents, each time a relevant doc is retrieved
 - Avoids interpolation, use of fixed recall levels
 - MAP for query collection is arithmetic ave.
 - Macro-averaging: each query counts equally
- R-precision
 - If have known (though perhaps incomplete) set of relevant documents of size Rel , then calculate precision of top Rel docs returned
 - Perfect system could score 1.0.

Announcements

- Consultations (Monday in-person, Thursday online) will run till the final exam.
- Ed forum will be monitored and answered till the final exam.
- Project marking starts shortly (aim to finish by the end of next week) - you will be notified.

Finally

MyExperience: due very soon

Please give some constructive feedback, **thank you !**

Thanks for taking COMP6714

Good luck for your exam